

INTERMEDIATE JAVASCRIPT

FORMATIVE ASSESSMENT

KM-03: Intermediate Programming Principles in JavaScript

ELECTRONIC LEARNER ASSESSMENT PAPER

Skills Programme	SP-220373: JavaScript Programmer
Module code	900219-000-00-KM-03
NQF level / credits	Level 4 / 9 credits
Assessment type	Electronic internal formative assessment
Duration	150 minutes
Total	80 marks
Assessment decision	Competent (C) / Not Yet Competent (NYC)

Learner details

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ID / learner number	8309175131082
Cohort / group	Group 01
Assessment date	Click or tap here to enter details
Electronic filename	KM03 Formative Assessment

How to complete and submit this assessment

- Download this file from Moodle and open it in Microsoft Word. Select Enable Editing if Word displays a protected-view message.
- Complete the yellow fields. Use the dropdowns for multiple-choice, True/False and both Mix-and-Match sets.
- Click inside every written-response field and type your answer. The fields expand as you type.
- The theoretical and scenario sections test explanation and application. Do not write JavaScript code.
- Answer independently and follow the approved assessment rules for assistance, references and academic integrity.
- Save regularly and rename the file: Surname_Name_KM03_Formative_Assessment.docx.
- Upload the completed .docx file to the correct Moodle submission link. Do not convert it to PDF.

Electronic assessment note

This assessment totals 80 marks. The assessor records the score and the final Competent (C) or Not Yet Competent (NYC) decision according to the provider-approved assessment rule. Complete every field before submission.

Assessment structure

Section	Assessment method	Marks	Electronic response method
A	Multiple-choice	10	Select one option per question.
B	True or False	10	Select T or F per statement.
C	Two Mix-and-Match sets	10	Select one letter per concept.
D	Four theoretical questions	20	Type responses in the expanding fields.
E	Two guided scenarios	30	Type responses under each guided part.
TOTAL		80	
SECTION A Multiple-choice questions			10 MARKS

Select the best answer from the dropdown below each question. Each question carries 1 mark.

1. Which statement best describes encapsulation in object-oriented programming? [1 mark]

- A. Grouping related data and behaviour together while controlling how they are used
- B. Repeating the same instructions in several files
- C. Changing every object into an array
- D. Displaying all variables in the browser console

Your answer: A

2. Which structure is most suitable for one product containing a code, name, price and stock status? [1 mark]

- A. A single number
- B. An object
- C. A Boolean
- D. A loop

Your answer: B

3. Why is an array of objects useful for a product catalogue? [1 mark]

- A. It prevents every runtime error
- B. It stores only one property for the complete catalogue
- C. It stores several structured product records in one list
- D. It automatically sends the products to an API

Your answer: C

4. What is the main purpose of a catch block? [1 mark]

- A. To create a new HTML element
- B. To validate every form field automatically
- C. To repeat a request until it succeeds
- D. To handle an error raised while the try block runs

Your answer: D

5. What is the main benefit of client-side form validation? [1 mark]

- A. It gives the user immediate feedback before the form is sent
- B. It makes server-side validation unnecessary
- C. It guarantees that the network is available
- D. It permanently stores the submitted information

Your answer: A

6. What does a breakpoint allow a developer to do? [1 mark]

- A. Delete the complete script
- B. Pause execution and inspect the current program state
- C. Convert an object into a video
- D. Submit a form without validation

Your answer: B

7. Which console method is especially useful for viewing an array of objects in rows and columns? [1 mark]

- A. console.clear()
- B. console.warn()
- C. console.table()
- D. console.time()

Your answer: C

8. Which DOM method selects an element by its unique id? [1 mark]

- A. createElement()
- B. appendChild()
- C. remove()
- D. getElementById()

Your answer: D

9. Which addition most improves the accessibility of a product demonstration video? [1 mark]

- A. Captions for spoken information
- B. Faster automatic playback
- C. Flashing transitions
- D. Removal of playback controls

Your answer: A

10. What does response.ok help a developer check after a fetch request? [1 mark]

- A. Whether the page contains a form
- B. Whether the HTTP response indicates success
- C. Whether an object has a method
- D. Whether an animation has finished

Your answer: B

SECTION B | True or False**10
MARKS**

Select T for True or F for False. Each statement carries 1 mark.

No.	Statement	Electronic answer
1	A class can act as a blueprint for creating objects with similar structure and behaviour.	T - True
2	An object property may contain another object as its value.	T - True
3	A finally block can run whether or not an error was caught.	T - True
4	Client-side validation removes the need to validate information on the server.	F - False
5	A logical error always causes JavaScript to display a syntax-error message.	F - False
6	The <code>textContent</code> property can change the text displayed by a selected DOM element.	T - True
7	Toggling a CSS class can support a simple show-and-hide effect.	T - True
8	Audio and video controls should be usable by keyboard users.	T - True
9	<code>fetch()</code> automatically treats every HTTP 404 response as a rejected promise.	F - False
10	A breakpoint can help a developer inspect variable values before the next instruction runs.	T - True

SECTION C | Mix and Match**10
MARKS**

Complete both sets. Match every concept in Column A with the correct explanation in Column B. Each item carries 1 mark.

Set 1: Objects, errors and quality controls

Column A: Concept	Electronic answer	Column B: Explanation options
1. Class	D	A. Prevents an empty supported form field from passing built-in browser validation.
2. Nested object	B	B. Stores a structured object as the value of another object property.
3. catch block	E	C. Pauses program execution at a chosen instruction for inspection.
4. required attribute	A	D. Acts as a blueprint for creating similar object instances.
5. Breakpoint	C	E. Responds to an exception raised while the related try block runs.

Set 2: Browser interaction and communication

Column A: Concept	Electronic answer	Column B: Explanation options
1. <code>querySelector()</code>	C	A. Provides text for spoken or important audio information in a video.
2. <code>classList.toggle()</code>	E	B. Displays structured information such as objects in a readable table.
3. Captions	A	C. Returns the first DOM element that matches a CSS selector.
4. <code>response.ok</code>	D	D. Indicates whether an HTTP response has a successful status.
5. <code>console.table()</code>	B	E. Adds a CSS class when absent and removes it when present.

SECTION D | Theoretical questions**20
MARKS**

Answer all four questions in clear sentences or well-structured points. Each question carries 5 marks. No code is required.

1. Explain five beginner object-oriented programming ideas: class, object instance, encapsulation, inheritance and polymorphism. Use plain language and show how the ideas help organise a workplace system. [5 marks]

Class: A class is a basic blueprint used to create objects.

Object instance: An object is a real example created from a class.

Encapsulation: Encapsulation keeps related information together and protects it.

Inheritance: Inheritance lets one class reuse features from another class.

Polymorphism: Polymorphism lets different objects do the same thing in different ways..

2. A store product record must include basic properties, supplier details, a method that creates a short product summary, and a history of stock movements. Explain how an object, a nested object, an object method and an array of objects could organise this information, then state one safe way to update a property. [5 marks]

An object stores the product details, a nested object stores supplier details, an object method creates the product summary, an array of objects stores stock history, and a property can be safely updated by using its property name..

3. Distinguish a syntax error, runtime error and logical error. Then explain the purpose of try, catch and finally, including why a clear custom error message may help the user or support team. [5 marks]

A syntax error is a coding mistake, a runtime error happens while the program is running, a logical error gives the wrong result, and try checks the code, catch handles errors, finally runs at the end, while a clear custom message helps users or support understand what went wrong..

4. A customer contact form sometimes accepts empty names, rejects valid email addresses and displays the wrong success message. Explain three validation actions and two debugging actions that would help the team correct the form. [5 marks]

Check that the name is not empty, check the email format, and make sure required fields are filled in; then use breakpoints to check the code and the console to find errors causing the wrong success message.

SECTION E | Guided workplace scenarios**30
MARKS**

Answer every guided part. Apply KM-03 concepts in plain language. Do not write JavaScript code.

Scenario 1: Furniture service-request page**Scenario context**

A Lewis Group service department is reviewing a browser page used to capture furniture repair requests. The page records customer and product details, checks required information, responds when the form is submitted, and adds an approved request card to the page without reloading it. A current defect causes the page to reload and makes the success message disappear. The team also wants a clear show-and-hide effect for status details. Answer in plain language. Do not write JavaScript code.

a. Explain how one service request could be represented as an object. Identify suitable main properties, one nested group and one useful object method. [3 marks]

A service request can be an object with customer, product and repair details, a nested customer-details group, and a method that creates a short repair request summary..

b. Recommend two validation controls for the request form and explain the feedback the user should receive. [2 marks]

Check that required fields are filled in and the email is valid, with a clear message telling the user what needs to be fixed.

c. Describe a four-step debugging process for the reload-and-disappearing-message defect. Include reproduction, evidence, inspection and verification. [4 marks]

Reproduce the problem,
collect evidence,
inspect the code to find the cause,
then fix it and test again to make sure it works..

d. Explain how the submit event and the DOM should work together to show an approved request card without reloading the page. [3 marks]

The submit event should stop the page from reloading, then the DOM can add the approved request card and show it on the page..

e. Recommend a simple show-and-hide effect for request status details and explain three accessibility or usability safeguards. [3 marks]

Use a button to show or hide the status, make it keyboard-friendly, use clear text, and make sure the information is easy to find and understand..

Scenario 2: Product catalogue and media page

Scenario context

A Lewis Group product catalogue page must request furniture information from an approved REST API, check the response, and display product cards in the browser. Each card may include an image and a short demonstration video. The page should show a gentle visual state while content loads and a clear fallback when the request or media fails. Answer in plain language. Do not write JavaScript code.

a. Explain how a REST API request can retrieve the catalogue data. Include the endpoint, HTTP method and returned data. [3 marks]

The page sends a GET request to the approved API endpoint, which returns the furniture product details needed to display the catalogue.

b. Explain two checks or fallback actions that should occur before the page uses the returned product data. [2 marks]

Check that the data was received correctly and, if it fails or is missing, show a clear error message instead of using it..

c. Explain how the DOM can display the returned products as cards without replacing unrelated page content. [2 marks]

Use the DOM to add the product cards to a specific section of the page, without changing the other information already there..

d. Recommend five multimedia practices for the product image and demonstration video, covering format, controls, alternatives, captions and failure handling. [5 marks]

Use common image/video formats, add play and volume controls, provide alternative text, add captions, and show a clear message if the media fails to load..

e. Describe a simple loading or reveal effect and explain three principles that keep the animation useful and accessible. [3 marks]

Use a simple fade-in effect, keep it quick, make it easy to see, and make sure it does not prevent people using the page with a keyboard or assistive tools..

Learner declaration and submission check

Learner declaration

I declare that the responses in this document are my own work, except for any approved accommodation recorded by the assessor. I understand that unauthorised assistance or copied responses may be handled under the provider assessment and academic-integrity policies.

Learner typed full name	Shane Davis
Date completed	Click or tap here to type
Final saved filename	KM03 Formative Assessment Learner Paper

Assessor use only

Section A score	Assessor enters score
Section B score	Assessor enters score
Section C score	Assessor enters score
Section D score	Assessor enters score
Section E score	Assessor enters score
Total score out of 80	Assessor enters score
Overall decision	Select C or NYC ▼

Assessor feedback and remediation action

Click or tap here to type your response. The field expands as you type.